LexiGuard live retrieval ablation

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24 fresh CUAD questions, 6 public contracts, live Neo4j Aura

CRAG restored context precision from 0.465 in the unprotected hybrid to 0.729, matching the vector baseline. Its faithfulness was 0.934 versus the baseline's 0.959. This test does not establish an overall gain over vector retrieval.

Ragas results

Variant	Faithfulness	Context precision	Factual F1
Vector baseline	0.959	0.729	0.515
Hybrid without links	0.918	0.465	0.400
Hybrid with links, no CRAG	0.941	0.465	0.380
Hybrid + CRAG, evidence guard	0.934	0.729	0.490

Scores range from 0 to 1 and are case-macro means. All 24 cases are complete across all four variants for both primary metrics and supplementary factual F1. Raw F1 can penalize citations missing from short reference spans.

Findings

The relevance node compares semantic-only and hybrid evidence before answering. It retained semantic evidence in 22 cases and selected hybrid evidence in 2. This addresses graph fragments displacing complete vector passages; it does not prove that graph retrieval adds value on this sample.

The normal API recovered the audit/site-inspection, one-year renewal with 30-day notice, and non-compete failures used during development. These are reported separately from the fresh cohort. A liability-cap case (F24) remained unretrieved and CRAG abstained after two rewrites; it remains in the headline results.

Literal reference coverage in context was 80.5% for vector, 80.2% for guarded CRAG and 59.2% for unprotected hybrid. This is normalized word 5-gram recall, not semantic accuracy. No scope or stored source-window violations were found.

Method and limitations

All variants used the same pinned generator, separate judge, embeddings, answer prompt and final budget: at most 3 excerpts and 4,500 characters. CRAG also inspects both alternatives (up to 9,000 characters) and can rewrite twice, so total inference cost is not equal across arms. Graph reads execute over verified TLS; offline graph replay is forbidden.

Generator: CodeCraft gpt-5.6-luna (gateway-reported identity). Judge: NVIDIA nvidia/nemotron-3-super-120b-a12b. Embeddings: gemini-embedding-001, normalized 768 dimensions. One run and six curated, positive-answer contracts are insufficient to establish generalization. The initial negative 26-case live pilot is retained in the package; its cohort is not pooled with these results.

Supporting files and checks

The full report, frozen protocols and questions, exact measurement source, per-query scores, answers, executed Cypher and returned rows, source audits and cached responses are included in the full reproducibility package. One missing JSON comma was repaired without changing any judgment; the original response is preserved. Current source passed 62 tests and the frontend build. Aura has 30 source records and 1,351 clauses; Render redeployment remains separate.

CUAD: Hendrycks, Burns, Chen and Ball, NeurIPS 2021. Original dataset repository: https://github.com/The-Atticus-Project/cuad. This is an independent application evaluation.